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The Relationship of Leaf Area and Foliage Biomass to Sapwood Conducting Area in Four Subalpine Forest Tree Species

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ABSTRACT. Leaf dry weight and area were strongly correlated with sapwood area measured at 1.37 m, but leaf area per unit sapwood area varied widely among species: 1.88 $m^2 \cdot cm^{-2}$ for subalpine fir, 0.72 for Engelmann spruce, 0.44 for lodgepole pine, and 0.19 for aspen. The leaf area : sapwood area relationship was nearly constant for different portions of the crown, although the relationship was erratic in subalpine fir. Thus the upper portion of a larger tree has the same leaf area : sapwood area ratio as the entire crown of a smaller tree. This supports the hypothesis that a physiological balance exists between conducting tissue and the water requirements of the shoot. FOREST SCI. 27:477–482.

ADDITIONAL KEY WORDS. *Picea engelmannii, Abies lasiocarpa, Pinus contorta, Populus tremuloides.*

KNOWLEDGE OF LEAF AREAS of plants can be useful in many ways, including estimation of plant productivity, study of gas exchange phenomena, and determination of canopy cover densities. Leaf area meters are available for use on broadleafed foliage either detached from the plant or attached to small branches. However, the lack of direct, nondestructive measurements of leaf area of entire large plants or of canopies continues to be a problem.

The research of Grier and Waring (1974), Snell and Brown (1978), and others suggests that foliage biomass of a woody plant and the cross-sectional area of conducting tissue in the stem are highly correlated. Furthermore, because of the close relationship between foliage surface area and dry weight, leaf or needle area is also highly correlated with sapwood area (Rogers and Hinckley 1979, Waring and others 1977). Thus measurements confined to the stem can be used to estimate foliage area for the whole plant.

Our studies of consumptive water use by forest trees and of the role of trees in the hydrologic cycle require leaf area data. Such data have generally been lacking for species in the subalpine forests of the central Rocky Mountains. An estimate of the vertical distribution of foliage also is needed for studying gas exchange phenomena for various portions of the canopy.

Correlations between leaf area and area of stem conducting tissue may be based upon a physiological balance between demand for water by the crown and the ability of the stem to conduct it. Thus it may be hypothesized that the leaf area and stem area relationship could be defined not only for a whole tree crown and the entire conducting tissue beneath that crown, but also for a portion of the crown and the stem conducting tissue at the base of that portion. The top one-

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third of a crown might have the same leaf area: sapwood area ratio as the top two-thirds and the entire crown. If so, then leaf area by section of the crown could be determined by knowing the area of stem conducting tissue at the base of that crown section. The study reported here investigated the relationships between leaf area and sapwood area and their physiological basis in four subalpine tree species in the central Rocky Mountains.

MATERIALS AND METHODS

The study was conducted at the Fraser Experimental Forest, located 8 km southwest of Fraser, Colorado. All sampling was done in August and September 1979, at elevations ranging from 2,750 to 3,050 m. Data were collected for each of the important tree species in the subalpine forest zone—Engelmann spruce (*Picea engelmannii* Parry), subalpine fir (*Abies lasiocarpa* (Hook.) Nutt.), lodgepole pine (*Pinus contorta* Dougl.), and aspen (*Populus tremuloides* Michx.) Trees within each species were selected from a wide range of physiographic situations, including dry ridges, moist sites near streams, and slopes of various aspects, so that effects of site differences would be maximized.

The sampling procedure used for each tree was fairly similar to the methods used by Grier and Logan (1977). After marking the trunk at breast height (1.37 m), the tree was felled in a direction minimizing crown damage. The live crown, defined as that portion of the trunk from the tree top to the point of attachment of the lowest branch having live needles or leaves, was divided into thirds. At the base of each one-third, at 1.37 m above the ground, and at a point 0.1 to 0.2 m from the ground, the outside diameter of the trunk was measured. Then at each of these locations a thin trunk section (1.0 to 1.5 cm) was cut using a chain saw, and the average outer and inner diameters of sapwood were determined. The outer sapwood diameter was measured at the cambium. In all species the sapwood was difficult to distinguish from heartwood. However, when fresh-cut trunk sections were held up to the sun, the sapwood transmitted much more light than the heartwood (for sections large enough to have heartwood), and the border of the more dense heartwood was sharply defined. The inner sapwood diameter was estimated using the border between the two light transmission zones.

Branches of each crown third were detached and placed into stacks. Stacked branches for some crown thirds were nearly 2 m^3 in volume. Since drying space was limited, subsampling was required to determine leaf or needle mass. Subsampling was done by visually separating matched pairs of branches from a stack into two groups. For the largest trees, stepwise separating reduced the final sample to one-sixteenth of the original foliage; the final fresh volume of small branches was 0.1 to 0.2 m^3. Sample selection was reversed periodically to avoid left-side right-side biases. For subsequent separating, piles were selected randomly, and when two workers were preparing foliage they traded positions periodically. Thus any errors associated with the subsampling procedure or any worker biases were probably distributed randomly among the samples.

The final foliage samples were dried in an insulated building (about 20 m^3 volume) equipped with electric heaters totaling 5 kilowatts. Small branches with needles or leaves attached were dried in paper bags for 1 to 5 days, after which twigs were separated and removed. Final dry weights of foliage were determined after a minimum of 48 hours at 70° to 75°C following detachment from twigs.

The ratio of leaf or needle area to dry weight was determined on a composite sample of fresh foliage collected from at least 10 trees of each species (roughly 0.5 m branch length per tree). Leaves or needles of all ages were represented. For 3 segments per branch, needle areas were determined by measuring circumference of at least 5 needles at cross sections located at 3 positions along the

TABLE 1. *Equations describing leaf dry weight (LDW) as a function of sapwood area (SA) at 1.37 m. Leaf dry weight is in grams, and sapwood area is in cm^2.*

Species	Equation	r^2	Sample size
Engelmann spruce	LDW = 72.0 (SA)	0.99	10
Subalpine fir	LDW = 212.2 (SA)	0.93	8
Lodgepole pine	LDW = 46.2 (SA)	0.95	9
Aspen	LDW = 17.3 (SA)	0.97	11

length of the needles and multiplying the mean circumference by needle length. Leaf areas for aspen were determined by dot grids. In all cases, areas represent total rather than projected areas.

The fresh leaf area : dry weight ratio for each species was as follows: Engelmann spruce, 9.996 m^2 kg^{-1}; subalpine fir, 8.859 m^2 kg^{-1}; lodgepole pine, 9.518 m^2 kg^{-1}; and aspen 10.97 m^2 kg^{-1}. No estimate of error is available for this ratio. The value for lodgepole pine is 30.2 percent higher than Running's (1979) value of 7.31 m^2 kg^{-1} obtained for dry foliage. A surface area shrinkage of 23.2 percent would account for this difference; such shrinkage during drying is quite reasonable.

RESULTS

Linear regression analyses were performed on untransformed leaf weights and sapwood areas and on logarithmic (base 10) transformations. The log transformation did not improve the coefficient of determination (r^2) substantially. Therefore, the relationships between leaf weight and area and sapwood area for whole trees presented here are based on untransformed data.

The relationships of leaf biomass to sapwood area are given in Table 1. Coefficients of determination were 0.93 or higher. Using the leaf area : leaf dry weight ratios given above, the leaf area of each tree was determined from the total foliage dry weight. Total leaf area was linearly related to sapwood area measured at 1.37 m above the ground for all species (Fig. 1).

The range of sapwood area was much lower for subalpine fir than for the other three species (Fig. 1), although trees exceeding 25 cm in diameter were sampled for all species. In fir, extensive heartwood development reduced the sapwood to a thin cylinder having a maximum cross-sectional area of 109 cm^2. A rather careful visual examination of the crowns of firs indicated that other trees having much larger diameters (greater than 50 cm) had crowns (and presumably leaf and sapwood areas) no larger than those sampled in this study. Perhaps environmental and genetic factors combine to limit the development of an extensive conducting tissue and crown leaf area in subalpine fir.

The relationship between leaf area and sapwood area for crown sections is shown in Figure 2. Three data points, connected by a line, are given for each tree. The point for the top one-third of the live crown has the lowest leaf and sapwood area values for a particular tree and is closest to the origin. The middle point has as its ordinate the leaf area from the upper two-thirds of the crown and as its abscissa the sapwood area at the base of that portion of the crown (e.g., one-third of the way from the crown base to the top of the tree). The top point represents the leaf area for the entire crown and the sapwood area at the base of the crown. Data were plotted on a logarithmic scale to facilitate comparisons among crown positions over the wide range of sizes.

Linear regression analyses of the data presented in Figure 2 indicated that the leaf area : sapwood area relationship did not depend upon the portion of the crown

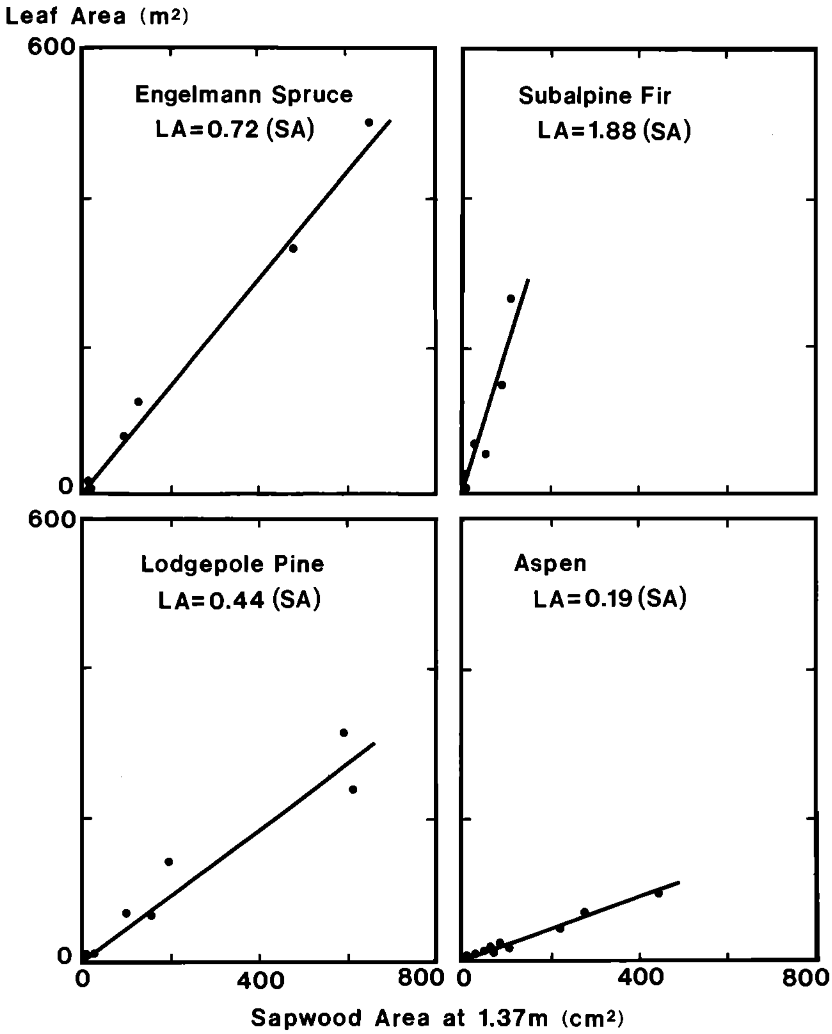

FIGURE 1. Relationship between leaf area (LA) for the entire crown and sapwood area (SA) measured 1.37 m above the ground for four subalpine forest tree species.

being studied. Stated differently, the top part of a larger tree had the same relationship as the entire crown of a smaller tree. Even though no significant differences occurred, four fir trees of the eight sampled showed a deflection to the right from the middle to the top points (Fig. 2). In these cases, the lower one-third of the crown added little leaf area, but sapwood area was considerably larger at the base of the total crown than it was at the base of the upper two-thirds of the crown.

DISCUSSION

From the slopes of the lines in Figure 1, the four species can be ranked in order of amount of leaf area served per unit of sapwood area. Values are 1.88 m^2 cm^{-2} for fir, 0.72 m^2 cm^{-2} for spruce, 0.44 m^2 cm^{-2} for pine, and 0.19 m^2 cm^{-2} for aspen. This ranking follows the relative tolerance of these species, with spruce and fir

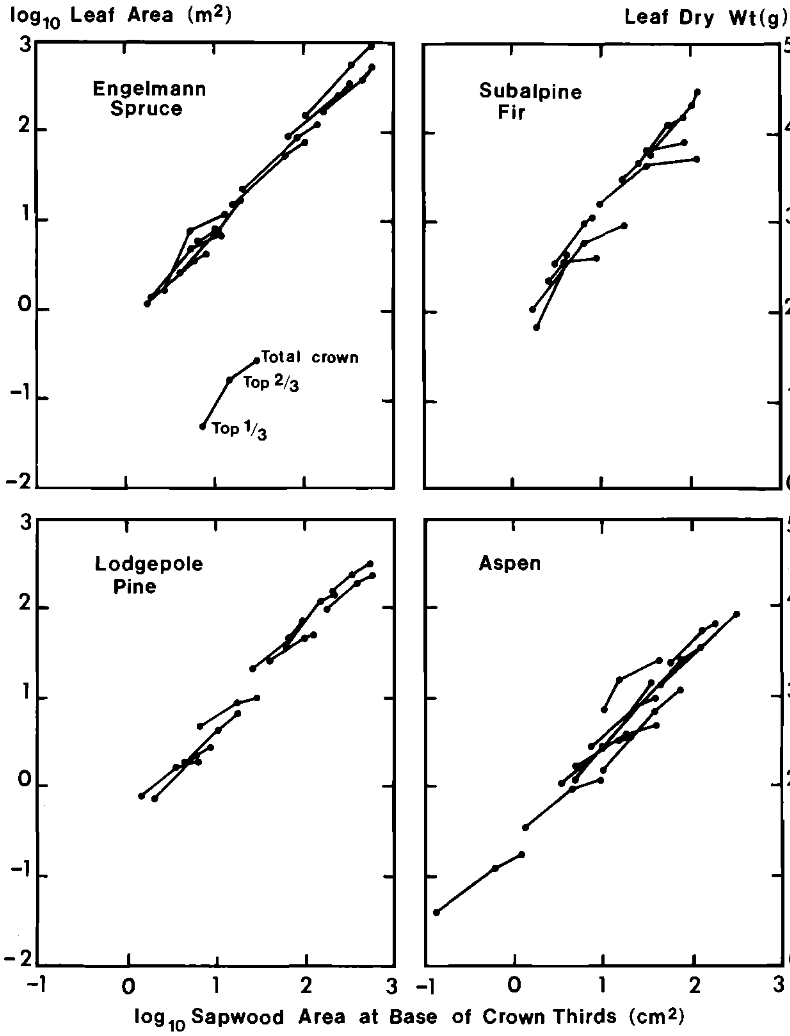

FIGURE 2. Relationship between leaf area and sapwood area for the total crown and the top two-thirds and top one-third of the crown. Lines connect the points for a single tree. See text for additional details.

being the most tolerant and aspen the least tolerant. Tolerant species generally are more protected from high radiation input and high evaporative demand than are intolerant species. Consequently, tolerant species may be able to carry higher leaf areas per sapwood area because of lower rates of water loss per unit leaf area. Interestingly, preliminary data describing the transpiration rates for each of these species appears to rank the species in the opposite direction, with aspen having the highest transpirational flux density (transpiration per unit leaf area per unit time) and subalpine fir the lowest under given environmental conditions. With a larger data base, it will become possible to calculate flux densities based upon sapwood areas and perhaps to evaluate the relative efficiencies of xylem transport among the species.

Gary (1976), in a detailed analysis of crown structure and biomass distribution for lodgepole pine in southern Wyoming, calculated the needle surface area for a 13 cm diameter tree. Assuming, from our observations on similar-sized trees, that bark thickness was 0.45 cm and that heartwood diameter was 5.5 cm, this tree would have 91 cm^2 sapwood area. From the equation in Figure 1, we estimate the surface area of needles for this tree to be 40.0 m^2. This value agrees well with Gary's calculated area of 42.5 m^2 obtained by somewhat different procedures.

While data are insufficient for careful prediction of leaf area, the relationships shown in Figures 1 and 2 indicate the feasibility of predicting leaf area for the whole crown or (with the possible exception of fir) for given layers or zones in the canopy. Research is continuing on the development of satisfactory predictive equations. Early indications are that external trunk measurements at 1.37 m (e.g., basal area) provide reasonable estimates of leaf area, with improvements when sapwood area or depth of crown are included.

Data presented here demonstrate a balance between leaf area and sapwood area which is contant among various positions in the crown. At the base of the crown, sapwood area is large because that area conducts water to all portions of the crown. Above the crown base, sapwood area is decreased in direct proportion to the reduction in leaf area.

During growth, plants acclimate to the environmental conditions to which they are exposed. A major developmental aspect of this acclimation is the size correlation among various plant parts. For example, plants which grow in regions characterized by poor soil water supplies generally develop larger root systems compared to shoots for the simple reason that large shoots would result in excessive water stress. Thus size of plant parts is coordinated through physiological feedback which in turn is influenced by conditions in the environment.

The leaf : sapwood area balance is maintained, presumably because physiological feedback remains constant throughout the crown. It is not known whether feedback works primarily in the direction of sapwood area controlling leaf area through limiting water supply, or in the direction of leaf area controlling sapwood area through limited carbohydrate supplies for growth. Nor is the role of physiological processes in heartwood formation understood. It may be useful to explore the relationships among leaf area growth, sapwood expansion (radial growth), and the progressive enlargement of heartwood in the context of physiological aspects of growth and water transport.

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